Experiment 5a(.sh file):

CODE:

```
  GNU nano 8.7.1
#!/bin/bash

read -p "Enter a string: " str

echo "Length of string: ${#str}"

read -p "Enter starting position: " start
read -p "Enter number of characters: " length

echo "Substring: ${str:$start:$length}"
```

OUTPUT:

```
tofiq@Tofiq:~$ bash exp5.sh
Enter a string: hello world
Length of string: 11
Enter starting position: 2
Enter number of characters: 5
Substring: llo w
```

Experiment 5b(.c file):

CODE:

```
  GNU nano 8.7.1
#include <stdio.h>

int main()
{

    char str[100];
    int start, length, i, count = 0;

    printf("Enter a string: ");
    fgets(str, 100, stdin);

    while (str[count] != '\0' && str[count] != '\n')
    {
        count++;
    }

    printf("Length of string: %d\n", count);

    printf("Enter starting position: ");
    scanf("%d", &start);
    printf("Enter number of characters: ");
    scanf("%d", &length);

    printf("Substirng: ");

    for (i = start; i < start+length && i < count; i++)
    {
        printf("%c", str[i]);
    }

    printf("\n");

    return 0;
}
```

OUTPUT:

```
tofiq@Tofiq:~$ ./exp5
Enter a string: hello world
Length of string: 11
Enter starting position: 2
Enter number of characters: 5
Substirng: llo w
```

Time Command:

```
tofiq@Tofiq:~$ time . exp5.sh
Enter a string: hello world
Length of string: 11
Enter starting position: 2
Enter number of characters: 5
Substring: llo w

real    0m10.271s
user    0m0.000s
sys     0m0.000s
```

```
tofiq@Tofiq:~$ time ./exp5
Enter a string: hello world
Length of string: 11
Enter starting position: 2
Enter number of characters: 5
Substirng: llo w

real    0m9.580s
user    0m0.003s
sys     0m0.004s
```